Advantages of Polymorphism:

- Flexibility and Extensibility: Allows adding new classes without modifying existing code.

- Code Reusability: One interface or method can be used for multiple object types.

- Easy Maintenance: Reduces class dependencies, simplifying code updates.

How Inheritance Supports Polymorphism in Java:

- Method Overriding: Subclass redefines a superclass method (e.g., Dog overrides Animal's makeSound()).

- Dynamic Dispatch: JVM determines the object's type at runtime to call the correct method (e.g., Animal animal = new Dog(); animal.makeSound(); calls Dog's method).

- Hierarchical Structure: Superclass provides a common interface for subclasses.

Differences Between Polymorphism and Inheritance in Java:

- Definition:

  +Polymorphism: Objects/methods take multiple forms based on context.

  +Inheritance: Subclass inherits properties/methods from a superclass.

- Purpose:

  + Polymorphism: Enables uniform handling of different objects.

  + Inheritance: Promotes code reuse and establishes class hierarchy.

- Types:

  + Polymorphism: Compile-time (method overloading) and runtime (method overriding).

  + Inheritance: Single, multilevel, hierarchical, or via interfaces.

- Implementation:

  + Polymorphism: Through method overriding or overloading.

  + Inheritance: Using extends for classes or implements for interfaces.

- Dependency:

  + Polymorphism: Often relies on inheritance for method overriding.

+ Inheritance: Can exist without polymorphism.